

Program : Diploma in Polymer Technology	
Course Code : 3092	Course Title: Polymer Science
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2,T:1,P:0)	Periods per semester: 45

Course Objectives:

- To explain the concept of Tg and crystallinity based on states of phases and aggregates
- To illustrate physical and chemical methods of polymer identification
- To provide concept of molecular weight and stereo regularity of polymers.
- To summarize polymer reactions and polymer modifications and its impact on environment

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic terminologies-monomer polymer, polymerisation chemistry and techniques, molecular weight of micromolecules and macromolecules		Fundamentals of polymer science	2
Basic chemistry		Applied chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain Tg , crystallinity and physical and chemical methods of polymer identification	10	Applying
CO2	Describe the concept of molecular weight and experimental method of its determination	12	Applying
CO3	Illustrate the stereo regularity of polymers	11	Understanding
CO4	Explain polymer reactions and polymer modifications	10	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain Tg, crystallinity and physical and chemical methods of polymer identification		
M1.01	Generalize concept Tg and crystallinity on the basis of States of phase and aggregates of polymers.	3	Understanding
M1.02	Explain Glass transition temperature (Tg)- Melting temperature (Tm) -Flow temperature (Tf)	2	Understanding
M1.03	Explain Crystalline polymer and amorphous polymer	2	Understanding
M1.04	Explain Polymer identification by chemical and physical methods	3	Applying
Contents: State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy.			
CO2	Explain the concept of molecular weight and experimental method of its determination		
M2.01	Interpret the Concept of molecular weight of micro and macro molecules.	2	Applying
M2.02	Explain different molecular weight of polymers, molecular weight distribution and polydispersity.	3	Applying
M2.03	Derive the mathematical expressions Mn& Mw	3	Understanding

M2.04	Summarize the various experimental methods for the determination of molecular weight of polymers	4	Understanding
	Series Test - I	1	
Contents: Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods.			
CO3	Illustrate the stereo regularity of polymers		
M3.01	Mention the microstructures based on chemical and geometrical structures	2	Understanding
M3.02	Explain Optical and geometrical isomerism in polymers	2	Understanding
M3.03	Comprehend the Importance of Ziegler Natta catalyst in stereoregularity of polymers	3	Understanding
M3.04	Explain the microstructures of PP,PB,CR,IR	4	Understanding
Contents: Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures.			
CO4	Explain Polymer reactions and polymer modifications		
M4.01	Identify the differences between reactions of polymers and simple molecules	1	Understanding
M4.02	Illustrate the effect of hydroxyl ,carboxylic and aldehydic groups in polymer Reactions	1	Understanding
M4.03	Explain acidolysis, hydrolysis, aminolysis addition and substitution reactions ,and crosslinking reactions in polymers	4	Understanding
M4.05	Summarize the physical and chemical methods of polymer modifications with applications	4	Understanding
	Series Test - II	1	
Contents: Differences between reactions of polymers and simple molecules - effect of functional			

groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting – IPN.

Text / Reference:

T/R	Book Title/Author
R1	Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New Delhi- 2005 Edition 3.
R2	Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, Singapore, 2008 Edition
R3	Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International (P) Ltd, New Delhi - 2005 Edition
R4	Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science International Ltd, 2005 Edition

Online Resources:

Sl.No	Website Link
1	https://books.google.com/books/about/Polymer_Science.html?id=mvCzE
2	libguides.uml.edu/polymer science